



SOUND-04

Section A

1. Due to the increase in length of pendulum in summer, its frequency period will [1]
 - a) Constant
 - b) Decrease
 - c) Remain unchanged
 - d) Increase
2. Which of the following method is used to reduce the noise pollution along roads? [1]
 - a) Planting trees around roads
 - b) Controlling traffic at road
 - c) Widening the road
 - d) Installing sound absorbent
3. Sound helps us to [1]
 - a) Group each other
 - b) Identify one another
 - c) View each other
 - d) Communicate with one another
4. If the thunder is heard by a man 4 seconds after the lightening is seen. How far is the lightening from man? [1]
(Speed of sound in air is 330m/s)
 - a) 1320 m
 - b) 1230 m
 - c) 1420 m
 - d) 330 m
5. The part which vibrates in our vocal-box is- [1]
 - a) Vocal cord
 - b) Air stem
 - c) Chord
 - d) Ear drum
6. The number of oscillations per second of a vibrating object is called its [1]
 - a) Amplitude
 - b) Frequency
 - c) Time period
 - d) Wavelength
7. A baby recognises her mother by her voice. The characteristics of sound involved is [1]
 - a) shrillness
 - b) loudness
 - c) quality
 - d) pitch
8. A child with hearing loss may have [1]
 - a) Defective speech
 - b) Longer tongue
 - c) Short stature
 - d) Black skinned
9. The frequency of a sound wave is 200 Hz, how many times will it vibrates in one minute? [1]

- a) 6,000 times
b) 200 times
c) 4,000 times
d) 12,000 times

10. Ultrasound equipment for investigating and tracking medical problems works at frequency **[1]**
a) Higher than 20 kHz
b) At 20 kHz only
c) Lower than 20 kHz
d) At any frequency

11. Which one of the following statements is correct regarding the velocity of sound? **[1]**
a) It is maximum in gases and minimum in liquids
b) It is maximum in solids and minimum in gases
c) It does not depend upon the nature of media
d) It is maximum in solids and minimum in liquids

12. A human heart beats 72 times in a minute. Its frequency will be **[1]**
a) 1.2 Hz
b) 8.2 Hz
c) 1.9 Hz
d) 2.5 Hz

13. Which of the following require material medium for propagation? **[1]**
a) Sound
b) Light
c) Transverse wave
d) All of these

14. Loudness of sound is measured in units of **[1]**
a) metre (m)
b) metre/second (m/s)
c) decibel (dB)
d) hertz (Hz)

15. The study of behaviours of vibrating string can be detected by of sound **[1]**
a) Sonometer
b) Hydrometer
c) Hygrometer
d) Barometer

16. When a tightly stretched band is plucked it **[1]**
a) Vibrate but do not produce sound
b) Vibrate and produce sound
c) Only oscillate to and fro
d) Produce sound but do not vibrate

17. Lightening and thunder occurs together but lightening is seen earlier than sound as **[1]**
a) The sound produced is inaudible
b) Sound travels in absence of light
c) Light do not require medium for propagation
d) Light travels faster than sound

18. A person rings a metallic bell near a strong concrete wall. He hears the echo after 0.3 s. If the sound moves with a speed of 340 m/s. The wall is far from him? **[1]**
a) 51 m
b) 11 m
c) 30 m
d) 102 m

19. Which of the following vibrates with a low frequency? **[1]**
a) Whistle
b) Sitar

- c) Flute d) Drum
20. In order to reduce the loudness of a sound we have to [1]
- a) increase its frequency of vibration of the sound. b) increase its amplitude of vibration of the sound.
- c) decrease its frequency of vibration of the sound. d) decrease its amplitude of vibration of the sound.
21. Which of the following quantities are proportional to each other? [1]
- a) Frequency and wavelength b) Amplitude and loudness
- c) All of these d) Time-period and frequency
22. For which measurement the 'decibel' is used? [1]
- a) Water vapour in air b) Haemoglobin in blood
- c) Sugar in urine d) Sound in environment
23. Vibration is also known as [1]
- a) Vibratory motion b) Oscillatory motion
- c) Periodic motion d) Non-periodic motion
24. A list of mediums is given below. [1]
- i. wood
- ii. water
- iii. air
- iv. vacuum
- In which of these mediums can sound travel?
- a) ii, iii & iv only b) i, ii & iii only
- c) iii & iv only d) i & ii only
25. Which of the following have voice of higher frequency and shriller? [1]
- a) Woman b) Man
- c) Dog d) Bat
26. Time taken by an object to complete one oscillation is called [1]
- a) Frequency b) Rarefaction
- c) Vibration d) Time period
27. Speed of sound in sea water is slightly more than distilled water because [1]
- a) Density of distilled water is more b) Density of sea water is less
- c) Sea is larger in size d) Density of sea water is more

Section B

28. **Assertion (A):** Vibration in a object makes it to produce sound. [1]
- Reason (R):** Vibration cause movement in air molecules.
- a) Both A and R are true and R is the correct b) Both A and R are true but R is not the

- explanation of A. correct explanation of A.
- c) A is true but R is false. d) A is false but R is true.
29. **Assertion (A):** Eardrum senses the vibration of sound and send signal to brain. [1]
Reason (R): When sound enters to ear, there is thin stretched membrane called eardrum.
- a) Both A and R are true and R is the correct explanation of A. b) Both A and R are true but R is not the correct explanation of A.
- c) A is true but R is false. d) A is false but R is true.
30. **Assertion (A):** Amplitude of sound decides it's loudness. [1]
Reason (R): Loudness of sound is proportional to square of amplitude.
- a) Both A and R are true and R is the correct explanation of A. b) Both A and R are true but R is not the correct explanation of A.
- c) A is true but R is false. d) A is false but R is true.
31. **Assertion (A):** In humans, sound is produced by vocal cords. [1]
Reason (R): When lungs force air through the slit, the vocal cords vibrate and produce sound.
- a) Both A and R are true and R is the correct explanation of A. b) Both A and R are true but R is not the correct explanation of A.
- c) A is true but R is false. d) A is false but R is true.
32. **Assertion (A):** Frequency of vibration is expressed in terms of hertz. [1]
Reason(R): The number of oscillations per second is called frequency of oscillation.
- a) Both A and R are true and R is the correct explanation of A. b) Both A and R are true but R is not the correct explanation of A.
- c) A is true but R is false. d) A is false but R is true.
33. **Assertion (A):** Sounds produced by horns of buses and trucks are considered to be pleasant sounds. [1]
Reason (R): Sounds which are pleasing to ear are called musical sound.
- a) Both A and R are true and R is the correct explanation of A. b) Both A and R are true but R is not the correct explanation of A.
- c) A is true but R is false. d) A is false but R is true.
34. **Assertion (A):** Sound does not need any medium to travel. [1]
Reason (R): Sound does not travel in vaccum.
- a) Both A and R are true and R is the correct explanation of A. b) Both A and R are true but R is not the correct explanation of A.
- c) A is true but R is false. d) A is false but R is true.
35. **Assertion (A):** Ultrasound equipment works on frequency higher than 20khz. [1]
Reason(R): The audible range for human ear is from 20 to 20khz.
- a) Both A and R are true and R is the correct explanation of A. b) Both A and R are true but R is not the correct explanation of A.
- c) A is true but R is false. d) A is false but R is true.

36. **Assertion (A):** A drum has low pitched sound than a whistle. [1]
Reason (R): pitch of sound depends upon loudness.
- a) Both A and R are true and R is the correct explanation of A. b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false. d) A is false but R is true.

37. **Assertion (A):** Plantations on the roadside can help to reduce noise pollution. [1]
Reason (R): Plantation will make the roads clean.
- a) Both A and R are true and R is the correct explanation of A. b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false. d) A is false but R is true.

Section C

38. **State True or False:** [23]
- (a) Speed of sound is greater than speed of light in air. [1]
 - (b) Auditory nerves transfer the signals to the brain. [1]
 - (c) Noise pollution may cause partial hearing impairment. [1]
 - (d) The ear drum acts as a stimulus that sensitises our ears. [1]
 - (e) Speed of sound is fastest in gases. [1]
 - (f) The number of oscillations per second of a vibrating object is called its time period. [1]
 - (g) SONAR stands for Sound Navigation and Ranging. [1]
 - (h) Sound of mosquito is produced by the vibration of its wings auditory nerves transfer. [1]
 - (i) Sound travels in vacuum. [1]
 - (j) If the amplitude of vibration is large, the sound is feeble. [1]
 - (k) For human ears, the audible range is 20 Hz to 20,000 Hz [1]
 - (l) Sound cannot travel in vacuum. [1]
 - (m) Repetition of sound due to reflection is called echo. [1]
 - (n) Unwanted and unpleasant sound is called music. [1]
 - (o) Outer ear collects the sound waves or vibrations. [1]
 - (p) Sound can travel through the vacuum. [1]
 - (q) Sound above 20 thousand hertz is known as ultrasound. [1]
 - (r) The loudness of a sound depend upon its frequency. [1]
 - (s) The time taken by an object to complete one vibration is called time period. [1]
 - (t) Sound waves require material medium to travel. [1]
 - (u) The lower the frequency of vibration, the higher is the pitch. [1]
 - (v) Larynx is sound producing organ in human. [1]
 - (w) Wind instruments produce sound by vibrating the air column. [1]

Section D

39. **Fill in the blanks:** [15]
- (a) The number of vibrations per second is called _____. [1]
 - (b) The loudness of normal breathing of human is _____. [1]
 - (c) Loudness of sound depend on _____. [1]
 - (d) Unpleasant sounds are called _____. [1]

- (e) Unwanted sound is called _____. [1]
- (f) Time taken to complete one oscillation is called _____. [1]
- (g) Frequency is measured in _____. [1]
- (h) The unit of frequency is _____. [1]
- (i) Time taken by an object to complete one oscillation is called _____. [1]
- (j) Shrillness of a sound is determined by the _____ of vibration. [1]
- (k) _____ is measured in hertz. [1]
- (l) Loudness is determined by the _____ of vibration. [1]
- (m) Above _____ the noise becomes physically painful. [1]
- (n) Excessive noise may cause many _____ related problems. [1]
- (o) Sound is produced when objects _____. [1]